

Reliability-Aware Fault Injection for Microscaling-Based DNNs

Problem Statement, Fault Model & Study Design — Working Draft

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Abstract

The adoption of microscaling formats in deep neural network (DNN) accelerators reduces the cost of low-precision inference, but their impact on DNN reliability under hardware faults remains largely unexplored. Unlike conventional floating-point representations, microscaling formats represent multiple values using a shared scale, making the effect of a bit fault dependent not only on the corrupted value but also on the scale and other values within the scaling block. We propose a reliability analysis methodology for microscaling-based DNNs that characterizes how faults in element values and shared scaling factors propagate through the network. The methodology evaluates fault sensitivity across different microscaling formats, block sizes, tensor types, and fault locations, and identifies the representation characteristics that make a model more vulnerable to faults. We conduct an extensive fault-injection study on representative CNN and Transformer models using weight and activation tensors. Our results show that reliability varies substantially across microscaling formats even when they provide comparable inference accuracy, and that faults affecting shared scaling information can have a significantly different impact from faults in individual elements.

Keywords — DNN reliability, microscaling formats, MXFP, block floating-point, fault injection, shared scale, fault sensitivity, low-precision inference.

1 Motivation and problem statement

Microscaling (MX) formats cut the cost of low-precision inference by having a block of K elements share one scaling factor: the stored value is $v_{i,s} = m_{i,s} \cdot 2^{(X_s - \text{bias})}$, where $m_{i,s}$ is a narrow 4–8-bit element and X_s is an 8-bit scale (E8M0) shared across the block. This coupling changes the fault model fundamentally. In conventional floating-point every value is a self-contained word, so a bit fault perturbs exactly one value. Under microscaling a fault's effect depends on the corrupted value, on the shared scale, and on the other values in the block — and it can occur at two structurally different sites:

SITE r — ELEMENT

Per-element value

A narrow 4/3/2-bit element m . A fault corrupts a single value.

blast radius $\rightarrow$ 1 value

SITE x — SHARED SCALE

Per-block shared scale

One 8-bit scale shared by K elements. A fault rescales the whole block.

blast radius $\rightarrow$ K values ($\times 2^{\Delta X}$)

Despite the rapid adoption of microscaling in DNN accelerators, its reliability under transient hardware faults is largely unexplored. This project provides a reliability-analysis methodology that characterizes how element and shared-scale faults propagate, and identifies which **representation characteristics** — format, block size, tensor type, and fault location — make a model more or less vulnerable.

CORE HYPOTHESIS

Reliability of microscaling DNNs varies substantially across formats **even at comparable inference accuracy**, and faults in the **shared scaling factor** behave very differently from faults in individual elements — because a single scale fault rescales an entire block of K values at once.

2 Objectives

- 01 Fault propagation.** Characterize how single-bit faults in element values vs. shared scaling factors propagate through the network, and quantify the difference in impact.
- 02 Format sensitivity.** Measure fault sensitivity across microscaling formats (MXFP8/6/4 and element variants) and test whether reliability differs even at comparable accuracy.
- 03 Block-size effect.** Determine how block size (scale-sharing granularity) trades inference efficiency against reliability.
- 04 Tensor & location.** Compare sensitivity across tensor types (weights vs. activations) and fault locations (scale bit vs. element bit; MSB vs. LSB; per layer).
- 05 Vulnerability drivers.** Identify the representation characteristics (dynamic range, exponent/mantissa split, scale granularity) that explain observed vulnerability.

3 Fault model

We adopt the single-bit-flip model standard in DNN fault-injection studies, generalized to the two microscaling sites, for both weight and activation tensors. Fault **occurrence** (where/how often a bit flips) is modeled separately from fault **impact** (how much the decoded value changes).

Element vs. shared-scale faults.

element flip (bit b of m): $d = |\log v' - \log v|$ — **1 value**
scale flip (bit b of X): $v'_{i,s} = m_{i,s} \cdot 2^{(X'-\text{bias})}$ — **all K values**
block severity $D_s = \sum_i |\log v' - \log v| = K \cdot |X' - X| \cdot \log 2$

A single shared-scale fault rescales the entire block by 2 raised to $(X'-X)$, giving it a blast radius of K — the structural reason scale faults differ from element faults.

Occurrence: per-position rate and multiplicity.

$$\pi_{\text{site},b} = \pi_{\text{site}}^0 \cdot \kappa_{\text{site}}(b) \quad P(S) = \prod_{b \in S} \pi_b \cdot \prod_{b \notin S} (1 - \pi_b) \\ H_{\text{site}} = 1 - \prod_b (1 - \pi_b) \approx \sum_b \pi_b + \sum_{b < b'} \pi_b \pi_{b'} + O(\pi^3)$$

$\kappa(b)$ = per-bit-position susceptibility profile (mean 1); a graded profile reproduces higher rates on more significant bits, $\kappa \equiv 1$ gives the uniform single-bit model. $P(S)$ = Poisson–binomial pattern probability; H = word-corruption rate, whose single-bit term is the baseline and multi-bit tail an optional extension.

4 Study design

The study sweeps four representation axes named in the objectives, over representative CNN and Transformer models and both tensor types:

FORMAT	MXFP8 (E4M3, E5M2), MXFP6 (E3M2, E2M3), MXFP4 (E2M1); optional MXINT8. Shared scale E8M0.
BLOCK SIZE	8 / 16 / 32 / 64 — scale-sharing granularity (fewer values per scale = smaller blast radius).
TENSOR TYPE	Weights (input-independent) and activations (input-dependent).
FAULT LOCATION	Shared-scale bit vs. element bit; MSB → LSB within each field; per-layer position.
MODELS	ResNet8, RepVGG-A0 (CIFAR-10); DeiT-Tiny/Small (ImageNet); optional small LLM.
FAULT MODEL	Single-bit flip at both sites; multi-bit (H-model) as an extension.
METRICS	Fault sensitivity = top-1 change / accuracy drop; per-site failure & SDC rate; per-bit / per-layer maps.

5 Methodology

Quantization. Quantize model weights and activations to each microscaling configuration; confirm baseline accuracy so reliability is compared at matched accuracy where possible.

Injection engine. Bit-addressable MXFP representation enabling faults at a chosen bit of either the shared scale or an element, for weights (static) and activations (per input, via hooks); golden-vs-faulty comparison detects top-1 changes and silent data corruption.

Campaign. For each cell of the study matrix, run a statistical fault-injection campaign at a target confidence/error. Value-aware stratified sampling (in the spirit of TreeFI, ICCAD 2026) is used as a supporting tool to keep the large multi-axis campaign tractable; it is not the contribution.

Ground truth. Exhaustive fault injection on a small model (ResNet8) where the microscaling bit space is small enough, providing a reference to validate the statistical estimates.

Analysis. Compare fault sensitivity across the four axes; contrast shared-scale vs. element faults; and correlate vulnerability with representation characteristics to explain the trends.

SUPPORTING TOOL, NOT THE CONTRIBUTION

TreeFI-style value-aware stratified sampling is referenced only to make the multi-axis campaign affordable. The contribution of this work is the **reliability characterization** of microscaling DNNs — not a faster injection method.

6 Expected outcomes

- A per-site reliability characterization showing shared-scale faults have a significantly different (typically larger) impact than element faults, consistent with their block-wide blast radius.
- Evidence that reliability varies substantially across microscaling formats even at comparable inference accuracy — so accuracy alone is not a reliability proxy.
- A block-size reliability trade-off: smaller blocks localize scale-fault damage at some storage/efficiency cost.
- Per-tensor and per-location sensitivity maps identifying which bits (scale vs. element; MSB vs. LSB) and which layers most need hardware protection.

- Design guidance: which microscaling configurations are most fault-robust for a given accuracy budget.

Open design choices (settled empirically): learned vs. exact interval characterization for the narrow element formats; single-bit vs. multi-bit (H-model) fault occurrence; and the block-size / format grid actually swept, bounded by available compute.